

SQL Part 2: COUNT, SUM, AVG, GROUP BY & HAVING Explained Simply — Free Practice Guide

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Step 1: Open a free online SQL playground

- sqliteonline.com — choose "SQLite", paste, click Run.
- db-fiddle.com — pick "SQLite" as the database, paste and Run.
- sqlime.org — a simple SQLite editor in your browser.
- sqlbolt.com — free interactive lessons for extra practice.

Step 2: Paste this setup script and click Run

It creates the same library tables used in the video. If anything goes wrong, just run it again in a fresh playground.

```
CREATE TABLE books (
  id INTEGER PRIMARY KEY,
  title TEXT,
  author TEXT,
  year INTEGER,
  genre TEXT,
  pages INTEGER
);
INSERT INTO books VALUES
  (1, 'Harry Potter', 'J.K. Rowling', 1997, 'Fantasy', 309),
  (2, 'Matilda', 'Roald Dahl', 1988, 'Comedy', 240),
  (3, 'Wonder', 'R.J. Palacio', 2012, 'Drama', 310),
  (4, 'The Hobbit', 'J.R.R. Tolkien', 1937, 'Fantasy', 310),
  (5, 'Holes', 'Louis Sachar', 1998, 'Mystery', 233),
  (6, 'Charlotte''s Web', 'E.B. White', 1952, 'Classic', 192),
  (7, 'Dog Man', 'Dav Pilkey', 2016, 'Comedy', 240),
  (8, 'Coraline', 'Neil Gaiman', 2002, 'Fantasy', 162);
```

Step 3: Run every query from the video, in order

Paste one at a time, click Run, and compare with the expected result.

COUNT: how many?

```
SELECT COUNT(*) FROM books;
```

```
Expected result:
COUNT(*)
8
```

SUM: add it all up

```
SELECT SUM(pages) FROM books;
```

```
Expected result:
SUM(pages)
1996
```

AVG: the average

```
SELECT AVG(pages) FROM books;
```

```
Expected result:
AVG(pages)
249.5
```

MIN and MAX

```
SELECT MIN(year), MAX(year) FROM books;
```

```
Expected result:
MIN(year) | MAX(year)
1937 | 2016
```

AS: friendly names

```
SELECT COUNT(*) AS total_books,        AVG(pages) AS avg_pages FROM books;
```

```
Expected result:
total_books | avg_pages
8 | 249.5
```

DISTINCT: no repeats

```
SELECT DISTINCT genre FROM books;
```

```
Expected result:
genre
Fantasy
Comedy
Drama
Mystery
Classic
```

GROUP BY: count per genre

```
SELECT genre, COUNT(*) AS books FROM books GROUP BY genre;
```

```
Expected result:
genre | books
Classic | 1
Comedy | 2
Drama | 1
Fantasy | 3
Mystery | 1
```

GROUP BY with AVG + ORDER BY

```
SELECT genre, AVG(pages) AS avg_pages FROM books GROUP BY genre ORDER BY avg_pages DESC;
```

```
Expected result:
genre | avg_pages
Drama | 310.0
Fantasy | 260.3333333333333
Comedy | 240.0
Mystery | 233.0
Classic | 192.0
```

HAVING in action

```
SELECT genre, COUNT(*) AS books FROM books GROUP BY genre HAVING COUNT(*) >= 2;
```

```
Expected result:
genre | books
```

```
Comedy | 2
Fantasy | 3
```

Putting it all together

```
SELECT genre, COUNT(*) AS books,         MAX(pages) AS longest FROM books WHERE year > 1950 GROUP BY genre HAVING COUNT(*) >= 2 ORDER BY books DESC;
```

```
Expected result:
genre | books | longest
Fantasy | 2 | 309
Comedy | 2 | 240
```

COUNT(DISTINCT)

```
SELECT COUNT(DISTINCT author) AS authors,        COUNT(*) AS books FROM books;
```

```
Expected result:
authors | books
8 | 8
```

WHERE + COUNT

```
SELECT COUNT(*) AS old_books FROM books WHERE year < 2000;
```

```
Expected result:
old_books
5
```

ROUND + top genre

```
SELECT genre, SUM(pages) AS total_pages,          ROUND(AVG(pages), 0) AS avg_rounded FROM books GROUP BY genre ORDER BY total_pages DESC LIMIT 1;
```

```
Expected result:
genre | total_pages | avg_rounded
Fantasy | 781 | 260.0
```

Step 4: Challenges (try first!)

C1. Show the average pages per genre, only for genres with 2 or more books.

C2. How many books were written before 2000?

C3. Total pages per genre, biggest first.

C4. How many different authors are there?

Answers

C1. Show the average pages per genre, only for genres with 2 or more books.

```
SELECT genre, AVG(pages) AS avg_pages FROM books
GROUP BY genre
HAVING COUNT(*) >= 2;
```

```
Expected result:
genre | avg_pages
Comedy | 240.0
Fantasy | 260.3333333333333
```

C2. How many books were written before 2000?

```
SELECT COUNT(*) AS old_books FROM books WHERE year < 2000;
```

```
Expected result:
old_books
5
```

C3. Total pages per genre, biggest first.

```
SELECT genre, SUM(pages) AS total_pages FROM books
GROUP BY genre ORDER BY total_pages DESC;
```

```
Expected result:
genre | total_pages
```

```
Fantasy | 781
Comedy | 480
Drama | 310
Mystery | 233
Classic | 192
```

C4. How many different authors are there?

```
SELECT COUNT(DISTINCT author) AS authors FROM books;
```

```
Expected result:
authors
8
```

Keep going!

Watch the whole free SQL course on Beyond School, starting with SQL Part 1: https://youtu.be/72JvYu2OtiA

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